

PRomop: A Longitudinal Decision-Ready Patient Health Record Built on OMOP CDM and FHIR R4

Adam Blum ¹

¹ HealthKey AI

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Software

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Summary

Health systems and biopharma face a persistent gap between holding patient data and acting on it. Records remain fragmented across providers, standards-conformant but manually mapped, and structured for storage rather than decision-making. Every downstream application — analytics, trial matching, clinical decision support — re-derives patient clinical state from scratch, multiplying effort and inconsistency.

PRomop is an open-source longitudinal patient health record built on the OMOP Common Data Model (CDM 5.4) ([OHDSI Collaborators, 2021](#)) with oncology extensions. Its keystone is PatientRecord, a flattened, denormalized projection that collapses each patient's complete longitudinal history into a single decision-ready row of over 300 columns. While the transactional CDM tables preserve everything that ever happened, PatientRecord represents what is true *now* — computing patient-state derivations once rather than repeatedly per consumer. Population analytics, clinical trial matching, and standard-of-care evaluation all operate on one shared substrate rather than maintaining divergent copies of the truth.

PRomop is deployed in production across the HealthTree Foundation (14,000 blood-cancer patients) and CancerBot (3,500 patients), supporting trial matching against 6,000 actively recruiting trials across five cancer types. Benchmarks show a $\sim 37\times$ speedup for eligibility screening compared to querying raw OMOP tables directly ([Blum et al., 2025](#)).

Statement of Need

PRomop is designed for clinical informaticists, data scientists, and developers who need to build or integrate with a longitudinal patient health record — whether to power a trial matching engine, construct feature sets for clinical ML models, or deploy patient-level clinical decision support.

Today, answering “what is this patient’s current disease status, most recent lab values, and prior therapy lines?” requires assembling and aggregating across multiple tables at query time. Every downstream application — analytics dashboards, trial matchers, CDS engines — independently reconstructs patient state: resolving lines of therapy, determining current disease status, normalizing biomarkers, reconciling conflicting source values. This re-derivation is expensive, error-prone, and a frequent source of inconsistency between applications that should agree. Neither OMOP CDM’s normalized tables nor FHIR’s resource-oriented exchange protocol (see *State of the Field*) produces a pre-computed, per-patient record ready for these use cases.

PRomop fills this gap by:

- Storing records in OMOP CDM 5.4, inheriting compatibility with the OHDSI ecosystem ([OHDSI Collaborators, 2021](#))

- Accepting FHIR R4 Bundle uploads that map directly into OMOP tables (observations → Measurement, conditions → ConditionOccurrence, medications → DrugExposure + Episode)
- Automatically deriving PatientRecord via a signal chain whenever any underlying OMOP record changes, so downstream consumers never reconstruct state themselves
- Exposing a versioned REST API (/api/v1/) with an OpenAPI 3.0 schema for integration with trial matching, CDS, and analytics services
- Providing a React-based clinician interface and synthetic FHIR data generators for each supported disease, enabling fully offline development and reproducible testing

State of the Field

Several tools address parts of the problem PRomop targets, but none combines OMOP-native storage, FHIR ingestion, and a pre-computed per-patient projection in a single open-source package.

OHDSI ATLAS / HADES / ACHILLES (OHDSI Collaborators, 2021; Overhage et al., 2012) are the reference tools for population-level observational research on OMOP CDM. They excel at cohort definition and epidemiological analysis but operate at the population level; they do not produce a queryable per-patient clinical state suitable for trial matching or point-of-care CDS.

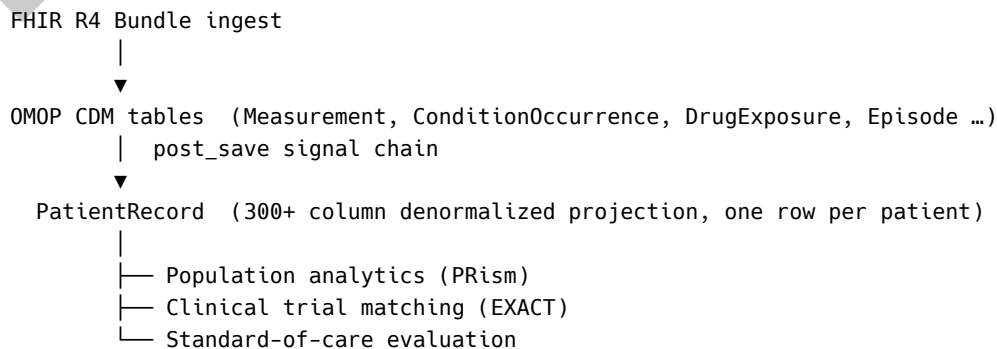
TrialGPT (Jin et al., 2024) applies large language models to match patients to trials from unstructured text. It bypasses structured data entirely, trading reproducibility and auditability for flexibility. PRomop takes the complementary approach: structured, vocabulary-mapped data with deterministic derivation.

HAPI FHIR (Smile CDR Inc., 2024) and other FHIR servers store and serve FHIR resources faithfully but do not map them into a common analytical schema. Querying current patient state still requires traversing and reconciling multiple resource types.

PRomop's contribution is the PatientRecord projection layer: a single denormalized row per patient, derived automatically from OMOP writes, that eliminates repeated state reconstruction across consuming applications.

Software Design

PRomop's central design trade-off is **normalization for writes versus denormalization for reads**. Clinical data arrives as FHIR R4 Bundles and is written into normalized OMOP CDM 5.4 tables (Measurement, ConditionOccurrence, DrugExposure, Episode), preserving full longitudinal history and OHDSI-ecosystem compatibility. A Django post_save signal chain then derives PatientRecord — a wide materialized projection (currently over 300 columns), one row per patient — on every write.



86 This design accepts higher write cost (the projection must be refreshed on each update) in
 87 exchange for dramatically lower read cost. A representative 20-criterion eligibility search over
 88 raw OMOP requires 27–39 joins; against PatientRecord it is a flat predicate over a single
 89 table. Benchmarks on a synthetic breast-cancer cohort measured a 37× speedup (0.30 ms
 90 vs. 11.0 ms per patient) for eligibility screening (Blum et al., 2025) (??).

Table 1: Eligibility screening cost per patient (synthetic breast-cancer cohort) (Blum et al., 2025).

Approach	Joins	Time per patient
Raw OMOP, 20 criteria	27–39	11.0 ms
PatientRecord, 20 criteria	0	0.30 ms
Measured speedup		~37×

91 Key implementation components include:

- 92 ■ **FHIR-to-OMOP ingestion** mapping LOINC, SNOMED CT, and RxNorm codes into
 93 OMOP tables with fallback to source value strings
- 94 ■ **Line-of-therapy inference** supplementing the ARTEMIS approach (Golozar et al., 2023)
 95 with domain-specific rules for oncology regimen detection
- 96 ■ **Synthetic data generators** producing reproducible FHIR R4 Bundles per disease type for
 97 fully offline development and testing
- 98 ■ **Versioned REST API** (/api/v1/) with OpenAPI 3.0 schema, OAuth2, and SMART on
 99 FHIR authorization (Mandel et al., 2016)

100 Research Impact Statement

101 PRomop is deployed in production across two independent organizations — the HealthTree
 102 Foundation (approximately 14,000 blood-cancer patients) and CancerBot (approximately 3,500
 103 patients) — totaling roughly 17,500 patient records. These deployments support clinical trial
 104 matching against 6,000 actively recruiting trials across five cancer types (multiple myeloma,
 105 follicular lymphoma, chronic lymphocytic leukemia, breast cancer, and diffuse large B-cell
 106 lymphoma).

107 The PatientRecord projection has enabled integration with two downstream systems: PRism,
 108 a population analytics dashboard, and EXACT, a clinical trial matching engine. Both consume
 109 the same pre-computed patient state rather than independently deriving it, eliminating a class
 110 of inconsistency bugs between applications that previously disagreed on patient status. A
 111 companion paper (Blum et al., 2025) provides architectural details and empirical benchmarks
 112 of the projection approach.

113 The software is openly available, includes synthetic FHIR data generators for each supported
 114 disease type, and can be deployed without access to real patient data — lowering the barrier
 115 for researchers and informaticists to adopt, extend, or benchmark against their own clinical
 116 data pipelines.

117 AI Usage Disclosure

118 Generative AI tools — primarily Anthropic Claude (via Claude Code) and GitHub Copilot —
 119 were used extensively throughout this project. AI assisted with software development (code
 120 generation, debugging, test authoring, and code review), documentation drafting (including
 121 portions of this paper), and architectural exploration. All AI-generated content was reviewed,
 122 tested, and edited by the author. Automated test suites (backend and frontend) were run
 123 against all code changes regardless of origin. The author accepts full responsibility for the
 124 correctness and scholarly integrity of the final software and manuscript.

